

9.5 EQUIVALENCE RELATIONS

A relation on a set A is called an equivalence relation if it is **reflexive**, **symmetric** and **transitive**.

Two elements a and b ^{that} are related by an equivalence relation are called **equivalent**.

The notation $a \sim b$ is often used to denote that a and b are equivalent elements with respect to particular equivalence relation.

For the notion of equivalent elements to make sense,

- Every element should be equivalent to itself, as the reflexive property guarantees for an equivalence relation.
- It makes sense to say that a and b are related by an equivalence relation, because when a is related to b , by symmetric property, b is related to a .
- As equivalence relation is transitive,
if a and b are equivalent and
 b and c are equivalent
it follows that
 a and c are equivalent

Example #1

Let R be the relation on the set of integers such that aRb iff $a=b$ or $a=-b$.

Sol: $R = \{(a, b) \mid a=b \text{ or } a=-b\}$ $A = \{\text{set of integers}\}$

Reflexive: $(a, a) \in R$ i.e.,

$a = a$ or $a = -a$, As we know that for every $a \in \mathbb{Z}$, $a = a$.

Symmetric: $(a, b) \in R \rightarrow (b, a) \in R$

As $(a, b) \in R$, then for integer a, b , we have $(b, a) \in R$

(i) $a=b$ $(2, 2) \in R \rightarrow (2, 2) \in R$

(ii) $a=-b$ $(2, -2) \in R \rightarrow (-2, 2) \in R$

$$\begin{array}{l|l} a=2 & a=-b \\ b=-2 & 2=-(-2) \end{array} \left| \begin{array}{l} a=-2 \\ b=2 \\ a=-b \\ -2=-(-2) \end{array} \right.$$

Transitive:

If $a=\pm b$ and $b=\pm c$, then $a=\pm c$

(i) $2=2$ & $2=2$ $2=2$

$\begin{array}{l} a=-b \\ b=-c \\ \hline a=-c \end{array}$ (ii) $2=-2$ $-2=2$ $2=2$

Hence relation R is Equivalence Relation.

Example #2

Let R be the relation on the set of real numbers such that
 $a R b$ iff $a - b \in \mathbb{Z}$.

Is R an equivalence relation?

Sol:-

Reflexive: $(a, a) \in R$

$a - a = 0 \in \mathbb{Z}$ for all $a \in \text{Real numbers}$
 Hence R is reflexive.

Symmetric: $(a, b) \in R \rightarrow (b, a) \in R$

if $a - b = K$, K is an integer, for all $a \neq b$ real
 then $-(a - b) = -K$
 $(b - a) = -K \in \mathbb{Z}$, for all $a \neq b$ real
 Hence, R is symmetric.

Transitive: $(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$

if $a - b \in \mathbb{Z}$
 and $b - c \in \mathbb{Z}$
 and we have to show that
 $a - c \in \mathbb{Z}$

By adding two integers, we obtain integer
 $(a - b) + (b - c) = \text{integer}$
 $a - \cancel{b} + \cancel{b} - c =$
 $a - c = \text{integer}$

Hence R is transitive.

$$\begin{array}{l} a = 2.5 \\ b = 1.5 \\ c = 4.5 \\ \hline a - b = 2.5 - 1.5 \\ \quad = 1 \in \mathbb{Z} \\ b - c = 1.5 - 4.5 \\ \quad = -3 \in \mathbb{Z} \\ \Rightarrow 2.5 - 4.5 = -2 \in \mathbb{Z} \end{array}$$

As R is reflexive, symmetric and transitive, So R is an EQUIVALENCE RELATION.

Example #3

Congruence Modulo m :

Let m be a positive integer with $m > 1$. Show that the relation

$$R = \{(a, b) \mid a \equiv b \pmod{m}\}$$

is an equivalence relation on the set of integers.

Sol:

Reflexive: $(a, a) \in R$

To show $a \equiv a \pmod{m}$

or $m \mid a - a$ i.e. $m \mid 0$ i.e. $\frac{0}{m}$ this is true for all positive integers m

So, the above R is reflexive.

Symmetric:

If we have $a \equiv b \pmod{m}$, then we have to show that $b \equiv a \pmod{m}$

as we know

$$a \equiv b \pmod{m} \Rightarrow m \mid a-b \Rightarrow \frac{a-b}{m} \Rightarrow a-b = m \cdot K \quad (K \in \mathbb{Z})$$

$$a-b = mK$$

$$\text{or } -(a-b) = -mK$$

$$(b-a) = (-K)m \Rightarrow m \mid b-a \quad \text{or } b \equiv a \pmod{m}$$

Hence R is Symmetric.

Transitive:

If we have

$$a \equiv b \pmod{m} \quad \text{and} \quad b \equiv c \pmod{m}$$

and to show that

$$a \equiv c \pmod{m}$$

$$\text{As } a \equiv b \pmod{m} \Rightarrow \frac{a-b}{m} \Rightarrow a-b = K_1 m \quad \text{--- ①}$$

$$\text{and } b \equiv c \pmod{m} \Rightarrow \frac{b-c}{m} \Rightarrow b-c = K_2 m \quad \text{--- ②}$$

Add ① and ②, we have

$$a-b + b-c = K_1 m + K_2 m$$

$$a-c = (K_1 + K_2) m$$

$$\Rightarrow m \mid a-c \quad \text{or } a \equiv c \pmod{m}$$

Hence R is transitive.

So, as R is reflexive, symmetric and transitive, then R is an equivalence relation.

Example #4

suppose R is the relation on the set of strings of English letters such that

$$a R b \text{ iff } l(a) = l(b), \text{ where } l(x) \text{ is the length of string } x$$

Is R is an Equivalence relation?

Sol:

Reflexive: $(a, a) \in R$

To show that $l(a) = l(a)$, obviously true ($a = \text{cat} : l(\text{cat}) = l(\text{cat})$
3 = 3)

Symmetric:

if we have $l(a) = l(b)$, then we have to show that $l(b) = l(a)$

As $l(a) = l(b)$ (length of a equals length of b)

then $l(b) = l(a)$ is also true

Transitive:

if we have $l(a) = l(b)$ and $l(b) = l(c)$, then definitely $l(a)$ will be equal to $l(c)$.

So, R is an Equivalence relation.

Example # 5

Let n be a positive integer and S a set of strings.

Suppose that $s R_n t$ iff $s=t$ or both s and t have at least n characters and first n characters of s and t are the same.

That is, a string of fewer than n characters is related only to itself; a string s with at least n characters is related to string t iff t has at least n characters and t begins with the n characters at the start of s .

For Example, let $n=3$ and let S be the set of all bit strings.

Then, $s R_3 t$ either when $s=t$ or both s and t are bit strings of length 3 or more that begin with the same three bits.

For instance,

$01 R_3 01$: string of fewer than n characters is related to itself;

$0011 R_3 00101$: a string s with at least n characters is related to a string t has n character and t begins with n characters at the start of s .

$01 \cancel{R_3} 010$

$01011 \cancel{R_3} 01110$

Show that for every set S of strings and every positive integer n , R_n is an equivalence relation on S .

Sol: $S = \{\text{set of all strings}\}$

Reflexive:

$s R_n s$: it is reflexive as here $s=s$ (where s is a string in S)

Symmetric:

If we have $s R_n t$, then we have to show that $t R_n s$

Let $n=3$

Case 1: $s=001, t=001$ ($s=t$ here)

we have $001 R_3 001$

Also we have $t R_3 s$ i.e. $001 R_3 001$

Case 2: $s=001101, t=001100$ (at least 3 characters begin with s and t)

we have

$s R_3 t \Rightarrow 001101 R_3 001100 \in R_3$

$t R_3 s \Rightarrow 001100 R_3 001101$ (at least $n=3$ characters same at the start of s and t)
 $\in R_3$

Hence, relation is symmetric

Transitive:

$$(s R_n t) \wedge (t R_n u) \rightarrow s R_n u$$

As we have

$s R_n t$: iff $s=t$ or both s and t have at least n same characters

$t R_n u$: iff $t=u$ or both t and u have at least n same characters

$n=3$

$s = 010111$

$t = 010101$

$$s R_3 t \Rightarrow 010111 R_3 010101 \in R$$

$$t R_3 u \Rightarrow 010101 R_3 010001 \in R$$

So,

$$s R_3 u \Rightarrow 010111 R_3 010001 \in R$$

As
: s and t both start with
 n characters

Hence R is transitive

So, R is an Equivalence Relation.

Example #6

Show that the "divides" relation on the set of positive integers is not Equivalence relation.

Sol:

Reflexive:

$a|a$: true for positive integers

Hence divide relation is reflexive.

Symmetric:

As we have $a|b$, we have to show that $b|a$ for all positive integers.

Suppose we have $a=2$, $b=4$

and $a|b$ i.e. $2|4$ ($\frac{4}{2}$)

But $b \nmid a$ i.e. $4 \nmid 2$ ($\frac{2}{4} = 0.5 \notin \mathbb{Z}^+$)

Hence, Divides relation is not symmetric, so

relation "divides" is not an equivalence relation.

Example #7

Let R be the relation on the set of real numbers such that

xRy iff x and y are real numbers that differ by less than 1, that is, $|x - y| < 1$

Show that ~~$|x - y| < 1$~~ R is not an equivalence relation.

Sol:

Reflexive: xRx

As $|x - x| = 0 < 1$ for all real values of x .

So, relation is reflexive

Symmetric:

if xRy , then yRx

As we have xRy

$$\text{i.e. } |x - y| < 1$$

where x and y are real numbers

which tells us ~~or~~ ^{that} $|y - x| < 1$

which tells us that

$$|x - y| = |y - x| < 1$$

So R is symmetric.

Transitive:

$$xRy \wedge yRz \rightarrow xRz \quad \text{for all real } x, y \text{ and } z.$$

Suppose $x = 2.8$, $y = 1.9$ and $z = 1.1$

As we have

$$xRy \quad \text{i.e., } |x - y| = |2.8 - 1.9| = 0.9 < 1 \quad f$$

$$\text{and } yRz \quad \text{i.e., } |y - z| = |1.9 - 1.1| = 0.8 < 1$$

$$\text{But } xRz \quad \text{i.e., } |x - z| = |2.8 - 1.1| = 1.7 \not< 1$$

So R is not transitive.

So, R is not an Equivalence Relation.

EQUIVALENCE CLASSES

Let R be an equivalence relation on a set A . The set of all elements that are related to an element a of A is called the equivalence class of a .

In other words, if R is an equivalence relation on a set A , the equivalence class of the element a is:

$$[a]_R = \{s \mid (a, s) \in R\}$$

If $b \in [a]_R$, then b is called a representative of this equivalence class.

Any element of a class can be used as a representative of this class. i.e., there is nothing special about the particular element chosen as the representative of the class.

Example # 8 :

What is the equivalence class of an integer for the equivalence relation given:

Example # 9 :

What are the equivalence classes of 0, 1, 2, 3 for Congruence modulo 4?

Sol:

The equivalence class of 0 contains all integers a such that

$$a \equiv 0 \pmod{4}$$

The equivalence class of 0 for this relation is :

$$[0] = \{\dots, -8, -4, 0, 4, 8, \dots\}$$

The equivalence class of 1 contains all integers a such that

$$a \equiv 1 \pmod{4}$$

The equivalence class of 1 for this relation is :

$$[1] = \{\dots, -7, -3, 1, 5, 9, \dots\}$$

Similarly,

$$[2] = \{\dots, -6, -2, 2, 6, 10, \dots\}$$

$$[3] = \{\dots, -5, -1, 3, 7, 11, \dots\}$$

Every integer is in exactly one of the four equivalence classes and that the integer n is in the class containing $n \bmod 4$.

Generalization: (replacing 4 with any positive integer m)

Congruence classes modulo m :

The Congruence classes ~~modulo m~~ of an integer a modulo m is

$$[a]_m = \{\dots, a-2m, a-m, a, a+m, a+2m, \dots\}$$

For instance,

$$[0]_4 = \{\dots, 0-2 \times 4, 0-4, 0, 0+4, 0+2 \times 4, \dots\}$$

$$= \{\dots, -8, -4, 0, 4, 8, \dots\}$$

$$[1]_4 = \{\dots, -7, -3, 1, 5, 9, \dots\}$$

$$[2]_4 = \{\dots, -6, -2, 2, 6, 10, \dots\}$$

$$[3]_4 = \{\dots, -5, -1, 3, 7, 11, \dots\}$$

Example #10 :

What is the equivalence class of the string 0111 with respect to the equivalence relation R_3 from example 5 on the set of all bit strings?

Sol :-

The bit strings equivalent to 0111 are the bit strings with at least 3 bits that begin with 011

$$[011]_{R_3} = \{ 011, 0110, 0111, 01100, 01101, 01110, 01111, \dots \}$$

EQUIVALENCE CLASSES AND PARTITIONS:

Example # 11

What are the sets in the partition of the integers arising from Congruence modulo 4?

Sol:

As we have seen in previous example, there are four Congruence classes, corresponding to $[0]_4$, $[1]_4$, $[2]_4$ & $[3]_4$.

They are the sets

$$[0]_4 = \{ \dots, -8, -4, 0, 4, 8, \dots \}$$

$$[1]_4 = \{ \dots, -7, -3, 1, 5, 9, \dots \}$$

$$[2]_4 = \{ \dots, -6, -2, 2, 6, 10, \dots \}$$

$$[3]_4 = \{ \dots, -5, -1, 3, 7, 11, \dots \}$$

These Congruence classes are disjoint and every integer is in exactly one of them.

In other words, these Congruence classes form a partition.

Partition of integers arising from Congruence modulo 5.

$$[0]_5 = \{ \dots, -10, -5, 0, 5, 10, \dots \}$$

$$[1]_5 = \{ \dots, -9, -4, 1, 6, 11, \dots \}$$

$$[2]_5 = \{ \dots, -8, -3, 2, 7, 12, \dots \}$$

$$[3]_5 = \{ \dots, -7, -2, 3, 8, 13, \dots \}$$

$$[4]_5 = \{ \dots, -6, -1, 4, 9, 14, \dots \}$$

Example # 12:

Let R_3 be the relation from example 5. What are the sets in the partitions of the set of all bit strings arising from the relation R_3 on the set of all bit strings?

Sol:

Note that every bit string of length less than 3 is equivalent only to itself. Hence $[x]_{R_3} = \{x\}$ ^{null}, $[0]_{R_3} = \{0\}$, $[1]_{R_3} = \{1\}$

$$[00]_{R_3} = \{00\}, [01]_{R_3} = \{01\}, [10]_{R_3} = \{10\}, [11]_{R_3} = \{11\}$$

Λ
0
1
00
01
10
11

Note that every bit string of length 3 or more is equivalent to one of the eight bit strings

000, 001, 010, 011, 100, 101, 110, 111

We have

$$[000]_{R_3} = \{000, 0000, 0001, 00000, 00001, 00010, 00011, \dots\}$$

$$[001]_{R_3} = \{001, 0010, 0011, 00100, 00101, 00110, 00111, \dots\}$$

$$[010]_{R_3} = \{010, 0100, 0101, 01000, 01001, 01010, 01011, \dots\}$$

$$[011]_{R_3} = \{011, 0110, 0111, 01100, 01101, 01110, 01111, \dots\}$$

$$[100]_{R_3} = \{100, 1000, 1001, 10000, 10001, 10010, 10011, \dots\}$$

$$[101]_{R_3} = \{101, 1010, 1011, 10100, 10101, 10110, 10111, \dots\}$$

$$[110]_{R_3} = \{110, 1100, 1101, 11000, 11001, 11010, 11011, \dots\}$$

$$[111]_{R_3} = \{111, 1110, 1111, 11100, 11101, 11110, 11111, \dots\}$$

These 15 equivalence classes are disjoint and every bit string is in exactly one of them.

These equivalence classes partition the set of all strings.

Set of All strings :

$$S = \{\lambda, 0, 1, 00, 01, 10, 11, 000, 001, 010, 011, 100, 101, 110, 111, 1000, 1001, 1010, 1011, 1100, 1101, 1110, 1111, \dots\}$$

$$P_0 \cup P_1 \cup P_2 \cup \dots \cup P_{15} = S$$

$$\& P_0 \cap P_2 \cap P_3 \dots \cap P_{15} = \phi$$

Example #13

List the ordered pairs in the equivalence relation R produced by the partition $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, $A_3 = \{6\}$ of $S = \{1, 2, 3, 4, 5, 6\}$

Sol:- partitions are $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, $A_3 = \{6\}$

To find the relation

$$A_1 \times A_1 \cup A_2 \times A_2 \cup A_3 \times A_3$$

$$= \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\} \cup \{(4,4), (4,5), (5,4), (5,5)\} \cup \{6,6\}$$

$$R = \{ \}$$

Example #13 :

Find the partitions of the set

$$S = \{a, b, c, d, e\}$$

and the relation $R = \{(a, a), (b, b), (c, c), (d, d), (e, e), (a, b), (b, a), (b, e), (e, b), (a, e), (e, a), (c, d), (d, c)\}$

Equivalent class of some element, say a in A is defined as:

$$[a] = \{x \in A \mid (x, a) \in R\}$$

Sol:

$$[a] = \{a, b, e\}$$

$$[b] = \{b, a, e\}$$

$$[c] = \{c, d\}$$

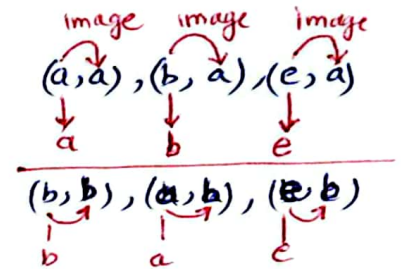
$$[d] = \{d, c\}$$

$$[e] = \{e, b, a\}$$

$$(c, c), (d, c)$$

$$(d, d), (c, d)$$

$$(e, e), (b, e), (a, e)$$



As these partitions are set (i.e. order does not matter),
So, there are only two partitions are here

Properties: $P_1 = \{a, b, e\}$ $P_2 = \{c, d\}$

① P_1 & P_2 are disjoint (i.e. $P_1 \cap P_2 = \emptyset$)

② $P_1 \cup P_2 = S$

③ $P_1 = \{a, b, e\}$: Here a, b and e are connected (they are related to each other in all possible manner)
 $P_2 = \{c, d\}$: Here c, d are connected

$$\Rightarrow P_1 \times P_1 \in R \quad P_2 \times P_2 \in R$$

$$(P_1 \times P_1) \cup (P_2 \times P_2) = R$$

• ~~Any~~ All partitions are mutually exclusive.